

ECON 609: Deep Learning

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Lecture # 8

What is Machine Learning?

- ▶ Wide set of algorithms to detect and learn from patterns in the data and use them for decision making or to forecast future realizations of random variables.
- ▶ Focus on recursive processing of information to improve performance over time.
- ▶ Think about the example of how to program a computer to play chess.

Why Now?

- ▶ Many of the ideas of machine learning are decades old.
 - ▶ Basic neural network by McCulloch and Pitts (1943)
- ▶ The availability of big data, cheap computational power, and algorithmic advances revived the ideas of machine learning.
- ▶ Many similarities with econometrics and statistical learning, but emphasis is somewhat different:
 - ▶ Practical algorithms vs. theoretical properties (scalability vs. asymptotic properties)
 - ▶ Traditional statistical inference is de-emphasized.
 - ▶ More interest in forecasting than in causality assertions.

Introduction to Deep Learning

- ▶ Machine learning is a catch-all name for a large family of methods.
- ▶ I will introduce deep learning. If you're interested in more ...
 - ▶ *Probabilistic Machine Learning* by Kevin Murphy
 - ▶ *The Nature of Statistical Learning Theory* by Vladimir N. Vapnik
 - ▶ *Artificial Intelligence – A Modern Approach* by Stuart Russell and Peter Norvig

The Problem

- ▶ We want to approximate ("learn") an unknown function:

$$y = f(x)$$

where y is a scalar and $x = \{x_0, x_1, \dots, x_N\}$ a vector.

- ▶ We care about the case when N is large (possibly in the thousands!).
- ▶ In economics, $f(x)$ can be a value function, a policy function, a pricing kernel, a conditional expectation, ...

A Neural Network

- ▶ An artificial neural network is an approximation to $f(x)$ for

$$y = f(x) \approx g^{NN}(x; \theta) = \theta_0 + \sum_{m=1}^M \theta_m \phi(z_m),$$

$\phi(\cdot)$ is an arbitrary activation function and $z_m = \sum_{n=0}^N \theta_{n,m} x_n$.

- ▶ The x_n 's are known as the features of the data.
- ▶ The $\phi(z_m)$'s are known as the representation of the data.
- ▶ M is the width of the model (wide vs. thin networks).
- ▶ "Training" the network: We select θ such that $g^{NN}(x; \theta)$ is as close to $f(x)$ as possible given some relevant metric.

Comparison with Projection Methods

- Compare a neural network with a standard projection.

$$(NN) : \quad f(x) \approx g^{NN}(x; \theta) = \theta_0 + \sum_{m=1}^M \theta_m \phi \left(\sum_{n=0}^N \theta_{n,m} x_n \right)$$

$$(CP) : \quad f(x) \approx g^{CP}(x; \theta) = \theta_0 + \sum_{m=1}^M \theta_m \phi_m(x)$$

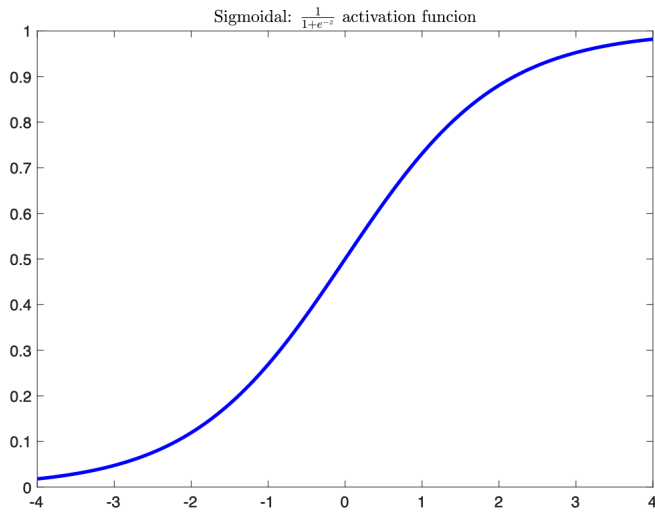
where ϕ_m is Chebyshev polynomial, for example.

- We exchange the rich parameterization of coefficients for the parsimony of basis functions.

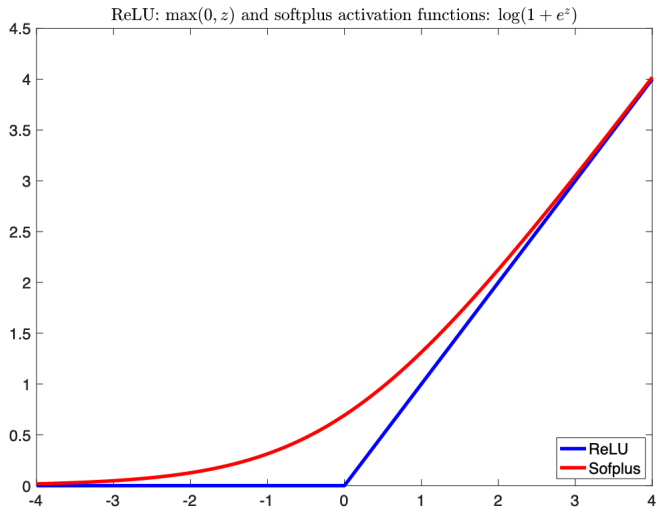
Activation Functions

- ▶ Traditionally, simpler functions as identity function $\phi(z) = z$ or a sigmoidal function $\phi(z) = \frac{1}{1+e^{-z}}$ were used.
- ▶ Some activation functions gained popularity recently:
 - ▶ Rectified linear unit (ReLU): $\phi(z) = \max(0, z)$.
 - ▶ Softplus: $\phi(z) = \log(1 + e^z)$.

Sigmoidal Function

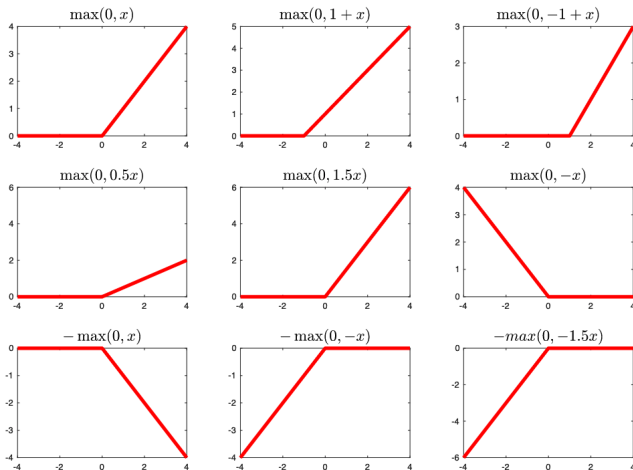


ReLU and Softplus



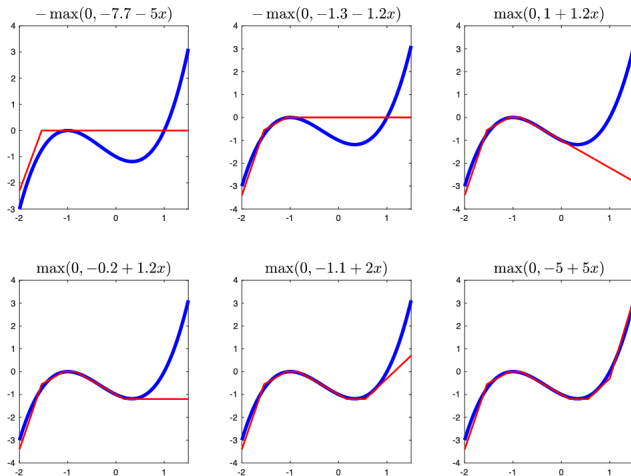
Shape of ReLUs

Different ReLUs: $\theta_i \max(0, \theta_{i,0} + \theta_{i,1}x)$



Approximation of $x^3 + x^2 - x - 1$

A six ReLUs approximation



- Example of a neural network with $M = 6, N = 1$.

Universal Approximation Theorem

- ▶ Hornik, Stinchcombe, and White (1989): A neural network with at least one layer can approximate any Borel measurable function mapping finite-dimensional spaces to any desired degree of accuracy.

Train the Network

- ▶ We need to specify a loss function to train the network (i.e. select θ).
- ▶ A natural loss function: the quadratic error function where

$$\theta^* = \operatorname{argmin}_{\theta} \frac{1}{2} \sum_{l=1}^L \|y_l - g(x_l; \theta)\|^2$$

- ▶ Where do the observations come from? Observed data vs. simulated data
- ▶ When solving macro models, Euler residuals can go into error function.

Why do Neural Networks Work?

- ▶ In a neural network, we take x , we perform affine transformations, and apply an activation function.
- ▶ In fact, a better name for neural networks could be *chained geometric transformations*.
- ▶ A deep learning network is a multilayer composition of neural networks.
- ▶ Deep neural networks look for convenient geometrical representations of high-dimensional manifolds.

Deep Learning

- ▶ A deep learning consists of layers of neural networks:

$$z_m^0 = \theta_{0,m}^0 + \sum_{n=1}^N \theta_{n,m}^0 x_n$$

$$z_m^1 = \theta_{0,m}^1 + \sum_{m=1}^{M^{(1)}} \theta_m^1 \phi^1(z_m^0)$$

...

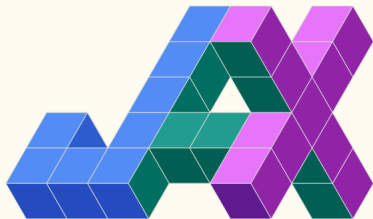
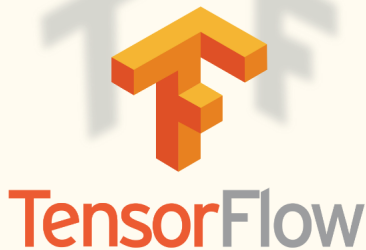
$$y \approx g^{DL}(x; \theta) = \theta_0^J + \sum_{m=1}^{M^{(J)}} \theta_m^J \phi^J(z_m^{J-1})$$

- ▶ $M^{(1)}, M^{(2)}, \dots$ and $\phi^1(\cdot), \phi^2(\cdot), \dots$ are possibly different across each layer of the network.

Why do Deep Neural Networks Work Better?

- ▶ Hidden layers induce highly nonlinear behavior in the joint creation of representations.
- ▶ Deep learning approach can efficiently approximate extremely complex functions.
- ▶ Open source libraries that integrate well with easy scripting languages.

ML libraries



PyTorch
Lightning



An Online Illustration

- ▶ We can play with many aspects of the network architecture with the right libraries.
- ▶ An interesting webpage: click [TensorFlow Playgrond](#)
- ▶ You can play with all the aspects of the architecture in several standard problems.

Limitations of Neural Networks and Deep Learning

- ▶ While neural networks and deep learning can work extremely well, there is no free lunch.
- ▶ We often require tens of thousands of observations to properly train a deep network. Real-life applications?
- ▶ If our goal is to forecast GDP next quarter, it is unlikely a deep neural network will beat an ARMA process...
- ▶ At least with solving dynamic models in economics, we can generate simulated data for long time periods.